



Asset Performance Management

PI Data Logger

User Guide

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INTRODUCTION

The PI Data Logger runs as a Windows service and will be deployed on Asset Performance Management application server. It retrieves tag data from PI historian via PI AF SDK. It generates data files for the Data Concentrator to process.

A Data Logger also generates heartbeat regularly by flipping a heartbeat attribute value when it can receive data from PI.

Installation: Install the software on Application Server.

PRE-REQUISITES

The following are the requirements for the Data Logger to function:

- Microsoft .Net framework v4.8
- Communication between DataBase and application servers.
- Download and install the PI AF Client 2018 SP3 (it can be downloaded from PI OSI software center).
- Trust connection enabled between the client and the PI server.
- Access to a Microsoft SQL Server database instances.

INSTALLATION

Follow the below steps for PI Data Logger installation:

1. In the installation media, double-click the **DataLoggerSetup.msi** file in the following folder:
Utilities\PIAFDataLogger

NOTE

On Windows 2012, the installer must be run as a user with permission to create directories.

2. **Data Logger Setup** dialog box appears.
Click **Next**.
3. Click **Browse** if you want to default install location.
Click **Next**.
4. **User Account Control** dialog box appears.
Click **Yes**.
5. After installation is completed, click **Close**.
6. Configure the Data Logger as described in the [Data Logger Configuration](#) chapter.
7. Save the configuration.
8. In the **Windows Search Box**, search and select **Services**.
9. In the Services panel, find **UAS PIAFDataLogger Service**, right-click and select **Properties**.
10. Change the service's user to the Asset Performance Management service account user.
11. Right-click on **UAS PIAFDataLogger Service** and select **Start**.
12. Synchronize the Data Logger with Data Concentrator as described in the [Synchronizing Data Logger with Data Concentrator](#) chapter.

NOTE

To check the service is working, open the **DCFileDropLocation** specified during Data Logger Configuration to view the generated CSV files.

NOTE

The following parameter values in **Honeywell.UAS.PIAFDataLogger.exe.config** and **Matrikon.DataLogger.Collector.Service.exe.config** need to be same.

- <piserverConfig PIServer="{PI Server name}"> and <datalogger ID="{PI Server name}">
- DCFileDropLocation and directory

DATA LOGGER CONFIGURATION

Follow below steps to configure a PI Data Logger:

1. Locate the following file:
[installdrive]:\Program Files
(x86)\Honeywell\MES\AssetManager\DataLogger\Honeywell.UAS.PI AFDataLogger.exe.config
2. Open **exe.config** file to view the additional application settings.
3. Update the connectionString to point to your SQL Server instance.
4. Locate **<appSettings>** section in **exe.config** file and update the **AMServiceURL**.
Example,
https://APMDITR532.DEV.LOCAL:443

NOTE

The AMServiceURL is the fully qualified domain name of web server.

5. Locate **<piserverConfigs>** section in **exe.config** file and update the following parameters:
 - a. **PIServer**
 - b. **DoDataSimulation**
 - c. **DCFileDropLocation**

If there are multiple PIServers, create copies of **<piserverConfigs>** section in the same **exe.config** file and update the above mentioned parameters.

Below are the list of parameters used in **exe.config** file:

Parameter	Description
AMServiceUrl	It is APM Web service endpoint's protocol and host name path.
QualifierSeparator	It is the CSV file name format.
RefreshConfigInMin	The time interval to check whether there are any changes for DL tag table in the DataBase .
PIServer	PI server to which Data logger is connected.
WorkLocation	The location where the service saves state information
DoDataSimulation	You can enable this to generate CSV file in case there are no PIServer
NOTE Set the value to false if there is a PIServer.	

Parameter	Description
SimDurationInSeconds	The data duration per data file.
SimDataRateInSeconds	The data rate per data file
HeartBeatFrequencyMinutes	The duration for which the value of heartbeat attribute needs to be updated.
DCQualifier	This qualifier is used to generate a part of CSV file name for a Data Logger.
DCFileTimestampFormat	The format of the file timestamp in .net date time format
DCDataTimestampFormat	The format of the data timestamp in .net date time format
DCFileDropLocation	The input file location of the configured DC data logger.
UseUTC	If the value is true , system will treat incoming data as UTC timestamped. If the value is false , system will treat the incoming data as local system time zone.
MaxRowsPerfile	The maximum number of rows that can be created per file.
DataRetentionMinutes	Retention time is the maximum duration the Data Logger retains the data before it's released for an output.
FileBatchSizeInSeconds	The data duration batch size of the data file.
MaxBackfillHours	The maximum time taken by the datalogger to go back to the backfill.
BackfillStepSizeInMinutes	The duration for a step size of backfill.
DistanceMinutesBetweenBackfillAndrealtime	It is the time difference between the backfill endtime and the snapshot latest timestamp.
FinalBackfillDelayMinutes	It is the delay to do final backfill after the subscription starts.
DiscardDataOlderThanXDays	Delete the Data Logger data from the server that is older than the user specified time interval. NOTE By default the data won't be deleted from the server unless number of days are specified.
DiscardOutOfSyncXHoursAgo	The Data Logger will discard the data that was out of sync specified hours ago.
MinimalDataFrequencyInSeconds	The minimal time interval taken by the Data Logger to collect inputs from the PI server.
DiscardFileLocation	The location where all the discarded files are saved.

NOTE

It is preferable to retain the default values for other parameters used in **<appSettings>** and **<piserverConfigs>** section. You can update the other parameter values if requirement.

SYNCHRONIZING DATA LOGGER WITH DATA CONCENTRATOR

The **Data Logger** need to be synchronized with the **Data Concentrator** for the service to work.

To synchronize Data Logger with Data Concentrator, follow the below steps:

1. Open any web browser.
Type `https://<machine name>:<port>/MES/Default.aspx#/UAS/UASLandingPage.aspx` in the address bar, where the **<machine name>** is fully qualified domain name of the Asset Performance Management Server and **<port>** is the configured SSL port.

The **Honeywell Application** landing page appears.

2. In the **System** area, click **System Configuration**.
3. Locate the following setting and update their values:

Key	Description
Honeywell.UAS.Data.DC.DataSourceNames	Name of PI server to which Data logger is connected. NOTE More than one PI server can be added by separating each PI Server name with a ",".
Honeywell.UAS.Data.DC.SyncConfigFromUAS	SyncConfigFromUAS enables the Data Concentrator to automatically synchronize with Data Logger. Update the key value to true .
Honeywell.UAS.Data.DC.SyncFromUAS	SyncFromUAS enables the Data Concentrator to automatically synchronize its tag definitions from APM. Update the key value to false .

4. Click **Save**.
5. Locate the following file:
`<installdrive>:\Program Files
(x86)\Honeywell\MES\AssetManager\DataConcentrator\DataConcentrator\Matrikon.DataLogger.Collector.Service.exe.config`
6. Open **exe.config** file to view the additional application settings.
7. Update the **QualifierSeperator** value.
The value specified in **Honeywell.UAS.PIAFDDataLogger.exe.config** and **Matrikon.DataLogger.Collector.Service.exe.config** need to be same.
8. Locate **<dataloggerConfigs>** section in **exe.config** file and update the following parameters:

Parameter	Description
ID	The PIServer value in Honeywell.UAS.PIAFDataLogger.exe.config and ID value in Matrikon.DataLogger.Collector.Service.exe.config need to be same.
templateOnly	Update the value to false .
directory	The DCFileDropLocation value in Honeywell.UAS.PIAFDataLogger.exe.config and directory value in Matrikon.DataLogger.Collector.Service.exe.config need to be same.

If there are multiple Data Source names in **System Configuration**, create copies of **<dataloggerConfigs>** section in the same **exe.config** file and update the above mentioned parameters.

9. Choose **Start > Run**, type **Services.msc** and click **OK**.

The **Services** window appears.

10. In the Services panel, find **Honeywell UAS Data Concentrator**, right-click and select **Restart**.

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